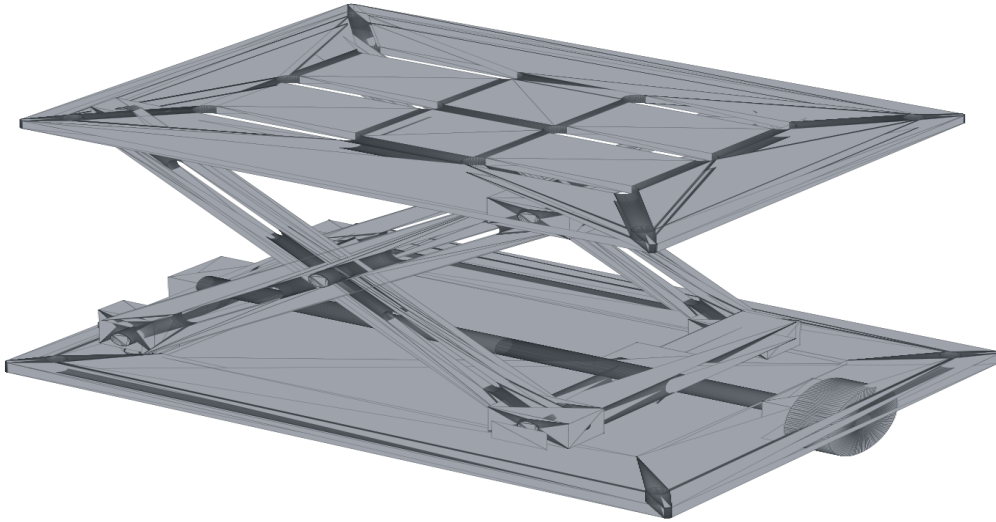


COMPACT SCISSOR LIFT MECHANISM

Self-Initiated Mechanical Design Project



Focus: Mechanical Design · CAD · Kinematics · Engineering Calculations · Technical Documentation

Project goal: Design a compact manual scissor-lift concept for a 10 kg centered payload and document the engineering process from requirements through CAD and analytical validation.

Design Approach

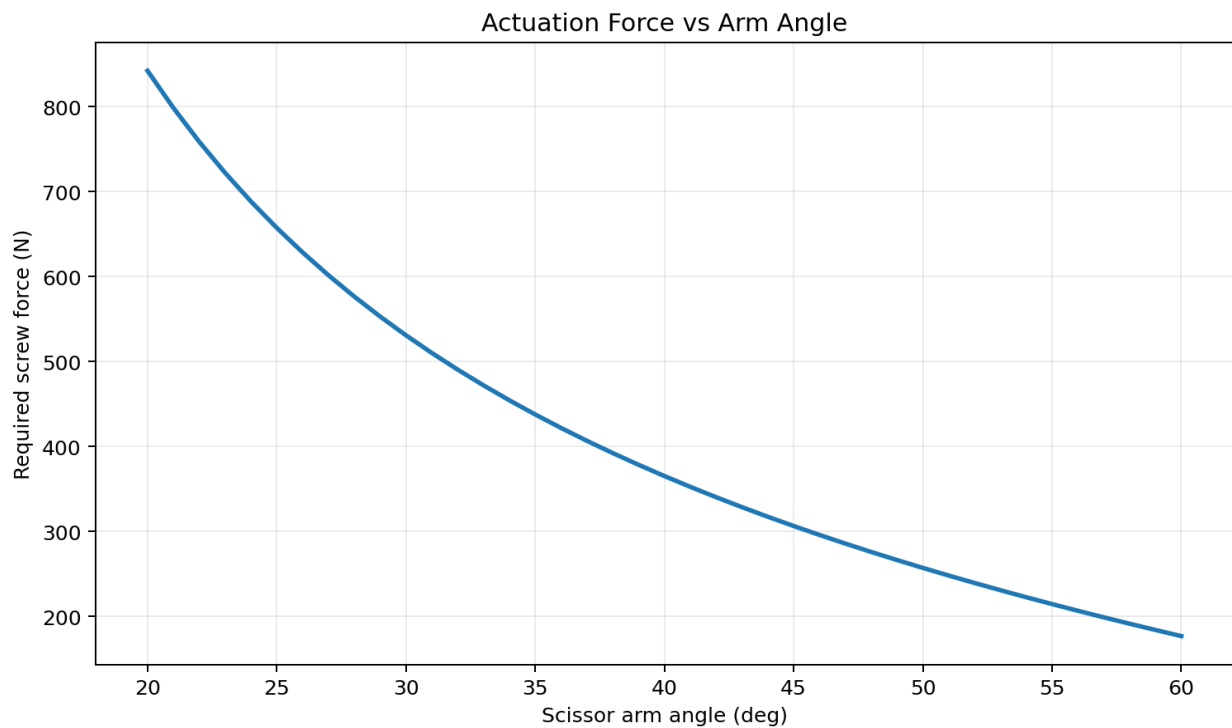
I structured the project around a clear engineering workflow: define requirements, establish scissor geometry, size the actuation concept, build the CAD assembly, check global stresses and deflection, document the BOM, and review design risks.



Key result: The selected 20°-60° operating range provides approximately 136 mm of linkage travel while avoiding the excessive actuator force associated with a near-collapsed scissor position.

Engineering Validation

The analytical design load was 306.6 N after including estimated moving mass, a dynamic factor, and a design safety factor. At the most demanding 20° position, the estimated screw force is 842 N and the lead-screw raising torque is about 1.08 N·m.



Global checks showed low arm axial stress and a high Euler buckling margin for the selected 30 x 6 mm scissor arms. A conservative strip model also estimated top-platform deflection at approximately 0.063 mm under the design load.

Project status: Virtual design complete. CAD, drawings, calculations, BOM, and design review are complete. Local FEA and physical prototype testing remain future validation steps.